

Multi-Cycle Verification and Testing

The screenshot below shows the registers updated by the RARS simulator after the RISC-V program ran.

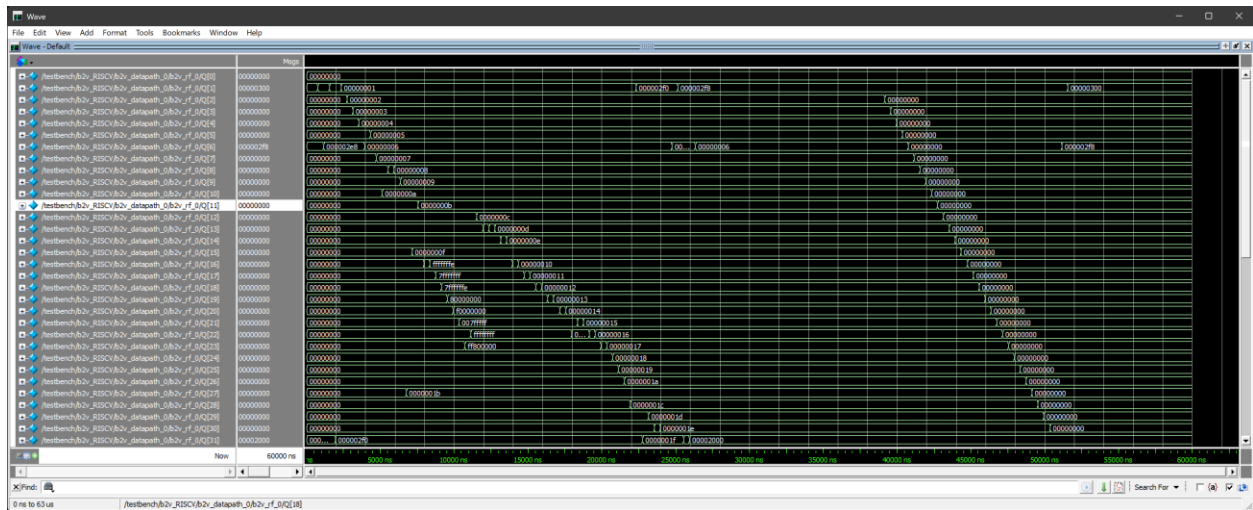
Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000300	
sp	2	0x00000000	
gp	3	0x00000000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x000002f8	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000000	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00002000	
pc		0x000001f0	

The screenshot below shows the RAM values of the RARS simulator after the program ran.

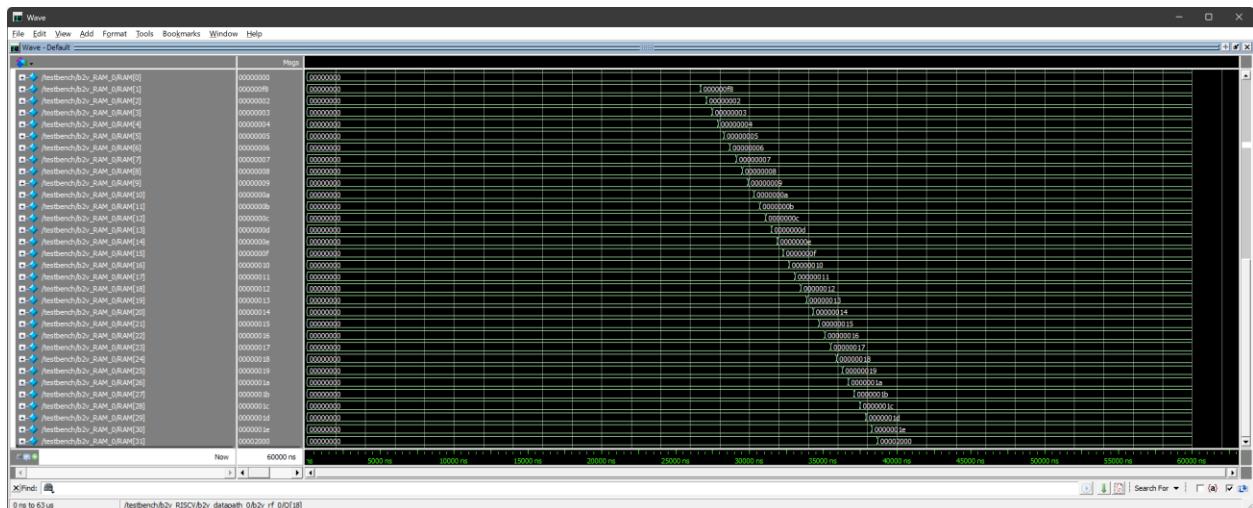
Data Segment								
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x00002000	0x00000000	0x000000f8	0x00000002	0x00000003	0x00000004	0x00000005	0x00000006	0x00000007
0x00002020	0x00000008	0x00000009	0x0000000a	0x0000000b	0x0000000c	0x0000000d	0x0000000e	0x0000000f
0x00002040	0x00000010	0x00000011	0x00000012	0x00000013	0x00000014	0x00000015	0x00000016	0x00000017
0x00002060	0x00000018	0x00000019	0x0000001a	0x0000001b	0x0000001c	0x0000001d	0x0000001e	0x00002000
0x00002080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000020a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000020c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000020e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00002100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00002120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00002140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00002160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00002180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000021a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000021c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x000021e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

0x00002000 (.data)
☒ Hexadecimal Addresses
 ☒ Hexadecimal Values
 ☐ ASCII

The screenshot below shows registers x0-x31 of the designed CPU in the ModelSim Wave window.



The screenshot below shows RAM locations 0-31 of the designed CPU in the ModelSim Wave window.



Comparing the register and RAM values and from the RARS simulator to the ModelSim simulation verifies that the CPU design correctly handles RISC-V instruction set.